

Project Design Phase

Proposed Solution Template

Date	15 February 2025
Team ID	HYD-METRO-2025
Project Name	Hyderabad Metro Rail – Digital Ticketing System (ServiceNow)
Maximum Marks	2 Marks

Proposed Solution Details

The project team has filled in the following proposed solution parameters for the Hyderabad Metro Rail Digital Ticketing System built on the ServiceNow platform.

1	Problem Statement (Problem to be solved)
Hyderabad Metro Rail commuters face daily frustration due to the absence of a digital ticketing platform. All ticket purchases are manual, counter-dependent, and cash-only. Commuters must stand in long queues every day, face fare calculation errors, and carry exact cash — resulting in missed trains, wasted time, and poor commuter experience. Station managers lack real-time visibility into booking data, making operational planning inefficient. There is no automated fare collection, no QR-based entry system, and no centralized digital record of passenger journeys.	
2	Idea / Solution Description
The proposed solution is a ServiceNow-powered Digital Metro Ticketing System that enables commuters to book metro tickets online through the ServiceNow Service Portal (/sp) — anytime, anywhere, from any browser-enabled device. Key components of the solution: • Catalog Item: 'Book A Metro Ticket' — a smart, dynamic booking form on the Service Portal • Automated Fare Calculation via Catalog Client Scripts (based on source, destination, journey type, and number of passengers) • QR Code Generation on form submission via GlideModal (no physical ticket needed) • Multi-mode Payment Support: UPI, Card, Wallet, Cash, or Others via catalog dropdown • Flow Designer Automation: post-submission data auto-mapped to Metro Database custom table • Business Rule: auto-generates ticket reference ID and triggers email confirmation • Role-Based ACLs: secure access to Metro Database (only managers/admins can view) • Service Portal Widget: QR display widget embedded on dedicated Metro Ticket portal page • Custom Tables: Metro Station Details (master data) + Metro Database (transaction records)	

3

Novelty / Uniqueness

- Built entirely on ServiceNow (no external development environment required) — leveraging a single enterprise platform for UI, logic, automation, data, and security
- Zero-infrastructure deployment: fully SaaS-based on ServiceNow cloud — no servers, no databases, no DevOps pipeline needed
- Dynamic QR generation within the platform using GlideModal + external QR API — no third-party ticketing software integrated
- Conditional UI logic (UI Policies) and real-time fare calculation (Client Scripts) create a native-app-like experience in a web portal
- Process automation via Flow Designer provides a no-code orchestration layer connecting catalog submissions directly to structured data records
- Extends ServiceNow beyond its traditional IT use case into public transport — demonstrating the platform's versatility for non-IT government services
- Replicable model: same architecture can be deployed for other metro cities (Bangalore, Chennai, Mumbai) with minor configuration changes

4

Social Impact / Customer Satisfaction

Social Impact:

- Reduces commuter queue time from 5–15 minutes daily to under 60 seconds per booking
- Promotes Digital India and cashless payment adoption among urban commuters
- Reduces paper token production and disposal — contributes to environmental sustainability
- Improves accessibility for working professionals, students, and persons with disabilities who cannot easily access physical counters
- Enables data-driven operational planning for metro authorities (peak hour analysis, route optimization, revenue tracking)

Customer Satisfaction:

- Commuters: Seamless booking experience with QR entry — eliminates the #1 pain point
- Station Managers: Real-time booking data visibility — no more manual register collection
- Metro Authority: Digital revenue tracking with audit trail in Metro Database
- IT Admins: Fully maintainable within ServiceNow — no custom code deployment needed
- Expected NPS improvement: from negative (frustration) to highly positive (delight)

5

Business Model (Revenue Model)

For Hyderabad Metro Rail Limited (HMRL):

- Primary Revenue: Ticket fare collection digitized — every booking creates a verified financial record in Metro Database, reducing revenue leakage from manual processes
- Operational Cost Reduction: Fewer counter staff needed as digital adoption increases; paper token production costs eliminated

- **Data Monetization (Future):** Anonymized passenger flow data can be used for retail space optimization, advertising targeting, and route planning contracts
- **Metro Card Recharge Revenue:** Digital recharge through the portal increases card usage frequency and float balance revenue for HMRL
- **Platform Licensing (Scalability):** The ServiceNow solution can be licensed/sold as a template to other Indian metro cities — creating a B2G revenue opportunity

For ServiceNow Implementation (Developer Perspective):

- Demonstrates ServiceNow's capability for public sector digital transformation projects
- Potential for paid implementation consulting engagements with metro authorities nationally

6

Scalability of the Solution

Technical Scalability (ServiceNow Platform):

- ServiceNow SaaS auto-scales horizontally — handles thousands of concurrent bookings without any infrastructure changes or performance degradation
- Custom tables (Metro Station Details, Metro Database) support millions of records with no schema modifications required
- Flow Designer processes submissions asynchronously — system never blocks the user experience during high-volume booking periods (peak hours)

Functional Scalability:

- **Station Coverage:** New metro stations added to Metro Station Details table reflect instantly in all catalog dropdowns — no code changes required
- **Route Expansion:** Additional fare routes added to the Client Script by updating route-fare mappings (future: dynamic fare matrix table eliminates even this)
- **City Expansion:** Solution deployable to any Indian metro city (Bangalore, Chennai, Kolkata) by importing update sets and updating station master data
- **Feature Expansion:** WhatsApp booking, mobile app, payment gateway integration, and analytics dashboards can be added as v2.0 modules without disrupting v1.0
- **User Scalability:** From a single PDI instance to enterprise multi-instance deployment with zero rearchitecting — only configuration changes needed